

NovaTech Sales Performance Analysis Report

1. Project Overview

This report analyzes the overall sales performance, profitability trends, regional distributions, product category breakdown, and discount strategy impacts for NovaTech. The analysis is based on the dataset of 600 transaction records processed through Python ETL pipeline, validated via PostgreSQL queries, and visualized in Power BI.

2. Dataset Summary

- Rows: 600 - Columns: 17 - Key Features:
 - **Transaction Identifiers & Dates:** Order ID, Order Date (Jan 2025 – Jun 2026)
 - **Geographics & Demographics:** Region (5 regions), State (18 states), Customer Segment (Consumer, Small Business, Corporate, Enterprise)
 - **Sales Channels & Staff:** Sales Channel (Online, Distributor, Direct Sales, Retail Store), Salesperson (10 sales reps)
 - **Product Information:** Category (6 categories), Product (18 tech products), Quantity, Unit Price (INR)
 - **Financial Metrics:** Discount (%), Gross Sales (INR), Discount Amount (INR), Net Sales (INR), Cost (INR), Profit (INR)
- **Missing Data:** 6 values in State column, 4 values in Customer Segment column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.

	Order ID	Order Date	Region	State	Customer Segment	Sales Channel	Salesperson	Category	Product	Quantity	Unit Price (INR)	Discount (%)	Gross Sales (INR)	Discount Amount (INR)	Net Sales (INR)	Cost (INR)
0	ORD-10002	2025-04-13	West	Goa	Small Business	Retail Store	Kiran	Accessories	Wireless Mouse	11	2500	10	27500	2750	24750	19990.84
1	ORD-10003	2026-01-21	South	Telangana	Small Business	Retail Store	Aarav	Monitors	NovaView 27	10	27100	5	271000	13550	257450	210245.56
2	ORD-10004	2026-05-20	North	Haryana	Consumer	Direct Sales	Priya	Monitors	NovaView 24	9	34100	2	306900	6138	300762	209995.67
3	ORD-10005	2026-01-23	South	Karnataka	Consumer	Direct Sales	Priya	Laptops	NovaBook Air	10	71600	15	716000	107400	608600	480128.66
4	ORD-10006	2025-12-04	East	Jharkhand	Consumer	Direct Sales	Aarav	Smartphones	Nova X1	7	66600	10	466200	46620	419580	321314.97

- **Data Cleaning:** Removed extra spaces from column names and text values.
- **Missing Values:** Filled missing State and Customer Segment values with "Unknown".

Order ID	0
Order Date	0
Region	0
State	0
Customer Segment	0
Sales Channel	0
Salesperson	0
Category	0
Product	0
Quantity	0
Unit Price (INR)	0
Discount (%)	0
Gross Sales (INR)	0
Discount Amount (INR)	0
Net Sales (INR)	0
Cost (INR)	0
Profit (INR)	0
dtype: int64	

- **Feature Engineering:** Created Year, Quarter, Month, and Month Number columns.
- **Profit Analysis:** Calculated Profit Margin % and Profit Per Unit.
- **Discount Analysis:** Created discount categories such as No, Low, Medium, and High Discount.
- **High-Value Orders:** Identified orders with sales of ₹5,00,000 or more.
- **Cleaned Data:** Saved the final data as NovaTech_Cleaned_Data.xlsx.
- **Database Integration:** Connected Python with PostgreSQL and loaded the cleaned data for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Total Sales and Profit** – Calculated the total sales revenue and total profit generated by the company.

	total_sales numeric	total_profit double precision
1	221408556	62341764.20000006

2. **Sales by Region** – Identified the region with the highest sales.

	Region text	total_sales numeric
1	South	64183234

3. **Revenue by Category** – Identified the category with the highest revenue.

	Category text	total_revenue numeric
1	Laptops	114352090

4. **Top Product** – Identified the best-performing product.

	Product text	total_sales numeric
1	NovaBook ...	49765480

5. **Sales by Salesperson** – Identified the salesperson with the highest sales.

	Salesperson text	total_sales numeric
1	Rohit	29957647

6. **Sales by Channel** – Compared sales across different channels.

	Sales Channel text	total_sales numeric
1	Online	70289406

7. **Monthly Sales Trend** – Analyzed sales changes month by month.

	month timestamp without time zone	total_sales numeric
1	2025-01-01 00:00:00	15442160
2	2025-02-01 00:00:00	12182377
3	2025-03-01 00:00:00	13267108
4	2025-04-01 00:00:00	14438518
5	2025-05-01 00:00:00	10915546
6	2025-06-01 00:00:00	8496204
7	2025-07-01 00:00:00	18305615
8	2025-08-01 00:00:00	11924619
9	2025-09-01 00:00:00	10897139
Total rows: 18		Query complete 00:00:00.062

8. **Profit Margin by Region** – Identified the region with the highest profit margin.

	Region text	total_profit double precision	total_sales numeric	profit_margin numeric
1	East	11740503.840000005	40645122	28.89

9. Low Profitability Products – Identified products with low profit margins.

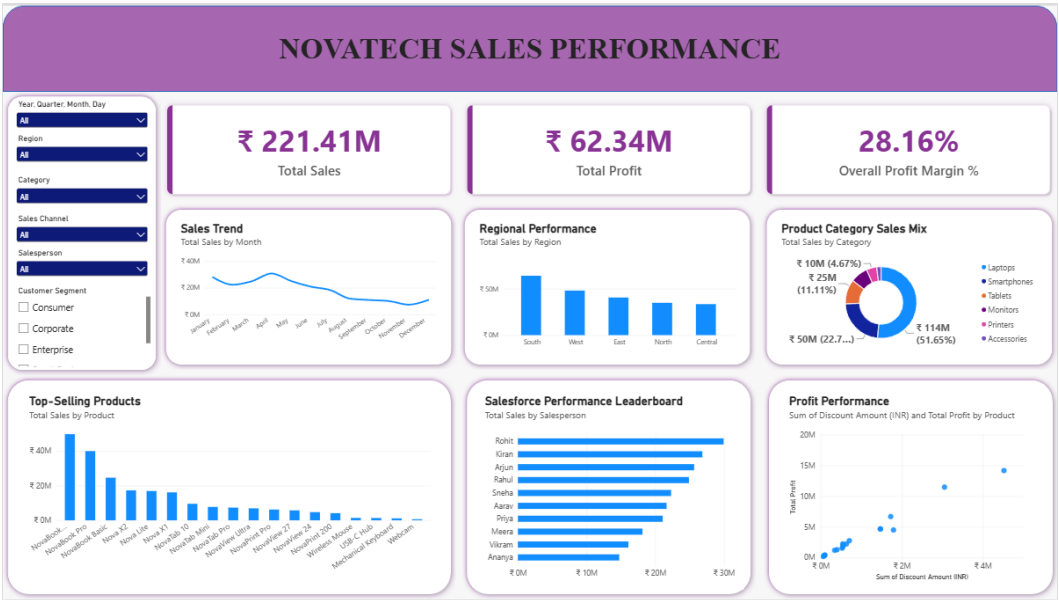
	Product text	total_sales numeric	total_profit double precision	profit_margin numeric
1	Webcam	613271	160159.71	26.12
2	NovaView 27	5723747	1529851.9000000001	26.73
3	Nova X2	17284631	4655528.05	26.93
4	NovaView 24	4710480	1269982.1300000004	26.96
5	NovaBook Basic	24611062	6677651.929999999	27.13
6	USB-C Hub	1328840	362474.30000000005	27.28
7	Nova Lite	16922769	4659487.480000001	27.53
8	Nova X1	16134299	4490976.14	27.83
9	NovaBook Air	49765480	14175596.690000005	28.48
Total rows: 18		Query complete 00:00:00.102		

10. Discount vs. Profit – Analyzed the relationship between discounts and profit.

	discount_level text	avg_discount numeric	avg_profit numeric	profit_margin numeric
1	Low Discount	1.13	112053.11	28.22
2	Medium Discou...	8.30	102215.32	28.10
3	High Discount	15.94	101655.47	28.49

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- **Focus on High-Margin Products:**
 - Promote Tablets (29.17% margin) and Laptops (51.65% revenue share).
 - Bundle low-margin Smartphones with high-margin Accessories.
- **Expand Key Regions:**
 - Grow sales in top markets: South (₹64.18M) and West (₹48.15M).
 - Boost marketing in underperforming markets: Central (₹33.49M) and North (₹34.94M).
- **Control Discounts:**
 - Keep standard discounts under 10% to protect profit margins.
 - Limit higher discounts (>10%) to bulk or clearance sales.
- **Reward Top Performers:**
 - Replicate sales strategies from top sellers like Rohit (₹29.96M).
 - Offer higher incentives for selling premium products (NovaBook Air, NovaBook Pro).
- **Prepare for Seasonal Peaks:**
 - Increase stock ahead of peak sales months (July and April).
 - Run special promotions during slow months (June and November).